EDAB6808 ASSIGNMENT 4

NLP of Bitcoin Tweets

Topic modelling and sentiment-volatility analysis

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Topic modelling on 4.69 million Bitcoin tweets identified ten themes, five of which are promotional templates rather than conversation, and sentiment scored on the same corpus does not improve next-day Bitcoin volatility forecasting.

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1. Data and preparation

The corpus is 4,689,305 tweets, not the 11,198,521 a default CSV reader reports. The file carries newline characters inside quoted tweet bodies. A standard reader splits one tweet across two or three rows, and every field after the break lands in the wrong column, which is why all thirteen columns are first typed as String, including follower counts and the retweet flag. The file also escapes internal quotes with a backslash rather than by doubling them, and some tweets contain backslashes as ordinary text. Therefore, no single parser setting handles both conventions.

When reading it with PyArrow, newlines in values and backslash escapes were expected; it parsed the file correctly and skipped 50 rows it could not resolve, a loss of 0.001%. Both row counts are reported here because counting rows off a default read would overstate the corpus by a factor of 2.4. After the correct read, 66 null dates and one non-numeric follower value remain out of 4.69 million, and the retweet flag is false on every row.

1.1 Cleaning decisions

Cleaning produced two text columns from one pass, because the two questions need opposite treatments. The topic column is lowercased and reduced to letters only. The sentiment column keeps case, punctuation, and emoji, all of which CryptoBERT and VADER read as signal.

URLs and mentions were removed from both, since a tracking link appearing in thousands of tweets would otherwise be one of the strongest features in the vocabulary. The hash symbol was stripped, but the hashtag word was kept, because #bitcoin and bitcoin are the same thing thematically. Digits were removed from the topic column only, since price figures fragment the vocabulary into thousands of single-use tokens, but retained for sentiment where the model reads them in context. Tweets under four words were dropped, and tweets over 30% non-ASCII were removed as a crude language proxy that catches CJK, Cyrillic and Arabic but not Spanish or French, which makes the corpus English-dominant rather than English-only.

Table 1. Attrition by cleaning rule.

Stage	Tweets	Cumulative loss
Raw	4,689,305	0.0%

Stage	Tweets	Cumulative loss
After null drop	4,689,239	0.0%
After minimum four words	4,615,967	1.6%
After non-ASCII cut	4,609,194	1.7%
After deduplication	2,956,373	37.0%

1.2 Duplication and automated accounts

The filters cost under 2% between them. Duplication removal accounted for the other 35.3%, which is 1.65 million tweets, and that is a finding rather than a cleaning step. The raw duplicate count on untouched text was only 119,155, or 2.5%. Stripping URLs and mentions exposed duplication hiding behind unique tracking links and rotating handles: the same tweet body posted thousands of times with a different link each time.

The top fifteen duplicated strings are price tickers and on-chain feeds without exception, led by an Ethereum price update appearing 30,815 times. Templates posted a hundred times or more number 1,886 and account for 823,596 rows. They collapse only because the cleaning strips the digits that made each one unique.

A second class of automated account survives deduplication, because a ticker posting a different price every ten minutes produces textually unique tweets. These were caught structurally: each tweet was reduced to its first five words, and accounts were scored on how many distinct openings they produce. Bitswings posts 9,707 tweets across 9 openings; Bitcoin Whale Alert posts 3,195 across one. The distribution is bimodal with almost nothing in the middle, the 5th percentile at 0.071 and the 10th at 0.400, and the threshold was placed in that gap at 0.15. Moving it from 0.05 to 0.30 changes the removal from 104,490 rows to 189,153, so the result does not depend on where in the trough it sits.

That leaves 2,794,803 tweets, from which 199,992 were drawn, stratified proportionally by month so that each month keeps its true share and the prevalence analysis is not carrying sampling noise. The sample is a compute constraint rather than a preference: embedding 2.79 million tweets and reducing them with UMAP exceeds the memory available in Colab. At this size, the standard error on a topic holding 5% share is under 0.1 percentage points, so the dominant themes are recovered essentially exactly. What is lost is the tail, where a theme appearing in 0.1% of tweets has 200 instances rather than 2,800 and may not survive as a cluster.

2. Question 1: method

Three representations were built on the same 199,992 tweets, because the three models take different inputs. TF-IDF with bigrams, capped at 20,000 features with a minimum document frequency of 20 and a maximum of 0.40, feeds NMF, which factorises a weighted term matrix and makes real use of phrase features. Raw counts feed LDA, whose generative model is about counts and which gains nothing from bigrams. Sentence embeddings from all-MiniLM-L6-v2, 384 dimensions, feed BERTopic, which clusters on meaning rather than shared vocabulary.

The words bitcoin, btc, and crypto were added to the stoplist. In a corpus of Bitcoin tweets, they appear almost everywhere and so cannot separate one topic from another; left in, they sit at the top of every topic and say nothing.

2.1 Selecting the number of topics

The number of topics was selected based on c_v coherence rather than chosen. NMF was fitted at ten values of k from 5 to 30 and scored on each.

Table 2. Coherence and reconstruction error by topic count.

k	NMF c_v	Reconstruction error
5	0.7629	441.49
8	0.8276	439.98
9	0.8454	439.50
10	0.8618	439.03
11	0.8541	438.64
12	0.8340	438.25
15	0.8006	437.12
20	0.7861	435.36
25	0.7747	433.84
30	0.7694	432.37

Coherence peaks cleanly at k=10 with 0.8618, and both neighbours sit below it, so the peak is not an artefact of where the grid landed. Reconstruction error runs the other way, falling monotonically from 441.49 to 432.37 with no turning point anywhere, because adding components always fits the matrix better. That is precisely why it cannot select k, and coherence can, and it is reported here as the contrast rather than as a second opinion.

LDA was swept over a narrower range around that peak and scored 0.5549 at k=8, 0.5510 at k=10, and 0.4824 at k=12. The gap between 8 and 10 is 0.004, which is nothing, so LDA is flat across that range and only tells us the answer is not 12. It does not contradict k=10, but it does not independently confirm it either.

The level gap between the two methods is the more useful result. NMF sits between 0.76 and 0.86 at every k tried; LDA never clears 0.56. LDA assumes each document is a mixture of several topics, which is reasonable for an article but wrong for a fifteen-word tweet about one thing. NMF makes no such assumption. We take k=10 from the NMF curve and carry LDA into the comparison as the contrast rather than as a candidate.

2.2 BERTopic as a third method

BERTopic determines its own cluster count from embedding density and found 132 topics, placing 95,040 tweets, 47.5% of the sample, in the outlier class. Reduced to ten for like-for-like comparison, it put 78,578 tweets into a single topic whose keywords match the outlier

class, because 132 narrow clusters merged into one when forced down to ten. The clusters that survived are events and campaigns rather than themes: BSC airdrops, NFT mints, Lightning faucets, NBA sportsbooks, the Tokyo Olympics, COVID vaccination, Publish0x referrals, and a token promotion.

The three methods are therefore answering different questions. NMF and LDA partition the whole corpus, and every tweet receives a topic, whether it has one or not. BERTopic claims only the half that clusters tightly and declines to classify the rest. The 47.5% outlier rate is the honest version of something NMF conceals: a large share of Bitcoin tweets are generic commentary belonging to no theme, and NMF assigns them to one regardless.

3. Question 1: the ten topics

Table 3. Final NMF topics at k=10, with labels assigned after reading keywords and representative tweets.

Label	Share	Leading keywords
Market speculation	49.4%	time, market, price, see, go, going, still, people, money, think
Crypto news and infrastructure	12.0%	ethereum, blockchain, cryptonews, trading, cryptocurrencies, altcoin
Airdrop promotion	8.4%	project, airdrop, good, airdrops, bsc, great, airdropinspector
NFTs	8.1%	nft, nfts, nftcommunity, opensea, nftart, metaverse, nftgiveaway
Altcoin tickers	7.7%	eth, doge, ada, xrp, shib, sol, bnb, solana, matic, dot
BSC yield farming promotion	5.6%	bnb, binance, bsc, defi, pancakeswap, yieldfarming
Trading talk	4.7%	buy, dip, sell, hodl, hold, bought
Lightning faucet referrals	1.9%	lightningnetwork, ln, lightning sats, faucet, boltcoiner, unlock
Publish0x referrals	1.3%	earn, post, read, referral, curated post, publish, articles
Coin Hunt World referrals	0.9%	coinhuntworld, playing, join, found, vault, vault location

Topic 3 holds 49.4% of the sample on its own. Its representative tweets are people speculating about where the price goes next, which is genuine Bitcoin discussion, but it is also everything that did not fit elsewhere. A topic containing half the data is a residual rather than a theme, and it is the weakest part of this decomposition.

The crypto news and altcoin ticker topics are arguably one theme split by vocabulary register rather than by subject. One spells out Ethereum, blockchain, and cryptonews as words and

news hashtags; the other is the same material in ticker form. The cosine similarity map between the NMF components is where that should be checked.

Five of the ten topics are not conversation at all. The Lightning faucet topic is the same sentence every time, with a different hex string in the middle. Coin Hunt World and Publish0x are both a fixed template with a username or article title swapped in. The BSC yield farming topic is the clearest case: identical promotional strings differing only in their opening words, variously “this will be huge”, “check one”, “very nice” and “good”. The airdrop topic is the same campaign at lower resolution. Together these account for 18.1% of the sample and occupy half the topic slots.

That is a limitation of the bot filtering, and it can be named precisely. The filter scores skeleton variety per account, which catches one account running a script ten thousand times but is blind to ten thousand accounts each posting a referral once, because every one of those accounts looks like an ordinary user. The duplication analysis showed exactly this pattern earlier, with one airdrop line posted 6,363 times across 6,219 distinct accounts. Deduplication missed the same tweets for the reason it missed the price tickers: each carries a unique hash, username or article title, so no two strings match.

Catching them would require filtering on the first eight tokens rather than the full string and dropping templates above a frequency threshold regardless of how many accounts posted them. That would free four or five topic slots and would probably break the residual topic into something more useful.

Prevalence was computed as monthly topic shares normalised within each month, so the chart shows composition rather than volume. That normalisation matters because tweet coverage is very uneven across the period, and apparent movement in a thin month is not comparable to movement in a well-covered one.

4. Question 2: measuring sentiment

The corpus changes between the two questions and the differences are deliberate. Question 1 removed the automated accounts because a ticker repeated ten thousand times becomes a topic. Question 2 retains them in the volume count, because a posting surge signals market attention whether a person or a script produced it. Sentiment itself is still scored on the deduplicated corpus, since scoring the tickers would drag every daily mean toward zero in proportion to how busy the bots were that day.

Scoring ran on a day-stratified sample capped at 1,500 tweets per day, giving 295,961 tweets across 201 days. At 1,500 draws, the standard error on a daily mean is around 0.02 on a score bounded in [-1, 1], far smaller than the day-to-day movement being measured, so the cap costs nothing. The 201 days matter more than the cap: the corpus spans just under twelve months but only 201 calendar days carry any tweets, so roughly 45% of the period is missing before the price merge.

4.1 Two methods

VADER is a rule-based lexicon designed for social media, extended here with sixteen crypto terms. Those weights are our own assumptions on VADER's scale, not measured values,

and they are needed because, out of the box, VADER scores bullish, rekt, and rugpull at zero. VADER produced 160,744 positive, 81,903 neutral, and 53,314 negative, with a mean compound of +0.237. Its 25th percentile is exactly 0.000, the signature of a lexicon method: a quarter of the corpus contains no word it recognises and scores zero by default rather than by judgement.

CryptoBERT is a transformer trained on crypto social media. It returns Bearish, Neutral and Bullish rather than negative, neutral and positive, which suits a market question but means the report uses its terms. Polarity was constructed as P(bullish) minus P(bearish), a signed score in [-1, 1] that behaves like VADER's compound. The label index order was read from the model configuration rather than assumed, because a wrong assumption would flip the sign of every score and invert the whole analysis with nothing in the output looking wrong.

CryptoBERT produced 147,742 bullish, 130,487 neutral, and 17,732 bearish, with a mean of +0.413 and a median of +0.480. Only 6.0% of the corpus reads as bearish, and three-quarters of all tweets score positive. That skew is a property of the model rather than of Bitcoin Twitter: it was trained on posts carrying user-applied Bullish and Bearish tags, where bullish dominates heavily.

The two methods correlate at only 0.145 and agree on 41.9% of labels. Two methods scoring the same tweets and agreeing on under half of them choose the sentiment method as a modelling decision rather than a detail. They are measuring different things: VADER measures whether the language is cheerful, CryptoBERT whether the poster expects the price to rise, and on a corpus this full of promotional enthusiasm those come apart badly.

4.2 Manual validation

Sixty tweets were labelled by hand, twenty from each predicted class so that all three categories are represented, with two raters working independently. The two raters agreed on 80% of the sample; Cohen's kappa was 0.686, which is substantial agreement on the Landis and Koch scale and a realistic figure for text this ambiguous.

Table 4. Accuracy against the manual sample, n = 60.

Measure	VADER	CryptoBERT
Overall accuracy	40.0%	48.3%
Bearish precision/recall	0.57 / 0.44	0.65 / 0.72
Bullish precision/recall	0.26 / 0.40	0.30 / 0.40
Neutral precision/recall	0.43 / 0.37	0.50 / 0.37

Neither method is good. Always guessing neutral scores 45% on this sample, so VADER barely clears the majority-class baseline and CryptoBERT beats it by three points. VADER's bullish precision of 0.26 means three out of four tweets it calls positive are not bullish, which is a general-purpose lexicon reading cheerfulness as market optimism. CryptoBERT's advantage is concentrated where it matters: bearish recall of 0.72 against 0.44 and bearish precision of 0.65 against 0.57, so it is substantially better at catching a negative market view. CryptoBERT was carried into the volatility modelling on that basis.

The tweets both methods got wrong are mostly promotional. “Go to do the needful and trust me you are worthy” with eight hashtags scores +0.87 on CryptoBERT, and a Crypto Expo Dubai countdown scores +0.72. Both read spam enthusiasm as bullishness, which is the Question 1 template problem reappearing as sentiment error. The genuinely hard cases were mixed time frames and dry irony rather than obvious sarcasm: a tweet reading “Now is the time for BUYING #Bitcoin, as you guys know, i predicted to drop from $44k to 38.500” is bullish about the present while describing a past fall, and VADER reads it as bearish off the word drop.

5. Question 2: volatility modelling

Garman-Klass was selected as the volatility target over absolute log returns and rolling standard deviation. Absolute return is a one-observation estimate of a daily standard deviation and is correspondingly noisy: its standard deviation of 0.0175 exceeds its own mean of 0.0164. Rolling standard deviation over five days is smoother, but the value at day t is built from days t-4 to t, so consecutive observations share four fifths of their inputs and the target becomes autocorrelated by construction rather than by anything about markets, which flatters any model fitted to it. Garman-Klass uses the open, high, low, and close of a single day, is roughly seven times more efficient than close-to-close, and being a one-day estimator, has no overlap between consecutive values. Its standard deviation of 0.0132 against a mean of 0.0209 is far better behaved. Rolling standard deviation and absolute return were retained as robustness checks.

The design is strictly lagged. The target is day t volatility, and the newest predictor is day t-1, so no information from the day being predicted enters the model. Market controls are yesterday's Garman-Klass, the trailing week and month averages as HAR terms, yesterday's return, yesterday's high-low range and logged market volume. Sentiment features are lagged daily mean polarity, its dispersion, the bullish-minus-bearish imbalance, the change in polarity and logged tweet volume. HAR terms are in the baseline deliberately: volatility is persistent, so an honest test of whether sentiment adds anything must beat volatility's own history.

Of 201 merged days, 179 survive the lags. Both series are stationary, with ADF p-values of 0.0000. The feature summary is itself a finding: lagged mean polarity has a standard deviation of 0.047 around a mean of 0.411, so it moves by under 12% of its own level across the entire period, and the bearish share averages 0.062 with a standard deviation of 0.023. Against that, lagged tweet volume has a mean of 15,282 and a standard deviation of 10,247, varying by two-thirds of its own size. The sentiment features are close to constant, and the volume feature is not.

5.1 Regression

Table 5. Market-only against market plus sentiment, HAC standard errors.

Specification	Adjusted R-squared	AIC	BIC
Market only	0.2625	-1076.3	-1054.0

Specification	Adjusted R-squared	AIC	BIC
Market plus sentiment	0.2642	-1072.0	-1033.8

Adding five sentiment variables lifts adjusted R-squared by 0.0017 and worsens both information criteria, because each penalises parameters that bought nothing. The joint F-test on the sentiment block returns p = 0.373. No individual sentiment coefficient is significant: mean polarity at p = 0.372, imbalance at p = 0.194, dispersion at p = 0.821, the polarity change at p = 0.294, and logged volume at p = 0.742.

What is significant is market structure. The lagged return carries a coefficient of -0.118 at p < 0.001 and the lagged high-low range 0.454 at p = 0.007, so yesterday's trading range and the direction of yesterday's move both predict today's volatility. The HAR terms are weaker than expected, with the monthly average at p = 0.063 and yesterday's volatility at p = 0.081, which on 179 observations is unsurprising.

5.2 GARCH

GARCH(1,1) was fitted as a second baseline rather than as the target, because it models conditional variance and does not observe it; using its fitted series as the thing being predicted would test whether sentiment predicts a model's output. The fit is weakly identified and should be reported as such: alpha of 0.163 at p = 0.334 and beta of 0.560 at p = 0.113 are both insignificant, so on 179 observations the model has not found a persistent variance process distinguishable from noise. The one strongly estimated parameter is the Student's t degrees of freedom at 3.12, p < 0.001, indicating extremely fat tails. The series is dominated by a few large moves, which is also why GARCH struggles to pin down the persistence terms.

The arch package has no GARCH-X: its exogenous argument enters the mean equation, not the variance equation, so placing sentiment in the conditional variance properly would require hand-coding the likelihood. The workaround used here is to take the fitted conditional volatility as a feature instead.

5.3 Rolling backtest

The backtest uses an expanding window, refitting at every step and predicting one day ahead, so the model only ever sees days before the one it predicts. With 120 days of burn-in, this leaves 59 forecasts, which is thin enough that differences of a few percent in error are not distinguishable from noise.

Table 6. One-day-ahead backtest, 59 forecasts.

Model	MAE	RMSE	SMAPE	Directional
OLS market	0.00986	0.01419	37.4%	51.7%
OLS plus sentiment	0.01020	0.01461	38.6%	44.8%
Random forest market	0.01014	0.01532	38.8%	48.3%
Random forest plus	0.01015	0.01505	39.1%	36.2%

Model	MAE	RMSE	SMAPE	Directional
sentiment				
XGBoost market	0.01109	0.01657	41.8%	48.3%
XGBoost plus sentiment	0.01036	0.01485	39.7%	51.7%

Read in pairs, sentiment does not help. OLS gets worse, random forest is unchanged, and XGBoost is the only pair that improves. It improves by moving toward the OLS baseline rather than past it, so the most plausible reading is that the extra features regularised a model that was overfitting on six, not that sentiment carries information.

Directional accuracy is the clearest column in the table. Every model sits at or below a coin flip at calling whether volatility rises or falls tomorrow, and the best two are the plain OLS baseline and XGBoost with sentiment, tied at 51.7%. Whatever the error columns show, none of these models has directional skill. The best single model is OLS on market variables alone, lowest on MAE, RMSE, and SMAPE and highest on correlation with the actual series at 0.386. A six-variable linear model beating both ensembles usually means the sample is too small for the ensembles to find anything a linear fit cannot.

5.4 High-volatility classification

The classification framing offered in the brief was run as an alternative. At the 80th percentile, the threshold is 0.02740 and 36 of 179 days are high-volatility. Market-only scores 49.2% accuracy and market plus sentiment 52.5%, a gain of three points on 59 test days, which is two days changing classification. The precision and recall split is more informative than the accuracy: both models recall the high-volatility class well, at 0.88 and 0.83, but with precision of 0.44 and 0.45, and both recall the normal class poorly, at 0.23 and 0.31. Balanced class weights have pushed them to flag aggressively, catching most volatile days by calling almost everything volatile. That is a model crying wolf, and precision rather than recall is the number to report.

6. Data integrity: the dates are off by four years

BERTopic surfaced topics about the Tokyo Olympics opening ceremony, COVID vaccination drives, NFT mints, and BSC airdrops. That is Bitcoin Twitter in 2021 and early 2022, while the date column reports March 2025 to March 2026. The column layout is also identical to a well-known Kaggle Bitcoin tweets dataset covering 2021 to 2023.

This was tested against date-anchored content, on the principle that Olympics tweets happen during the Olympics.

Table 7. Date probes against known real-world timing.

Probe	Tweets	Concentration	Real-world date
Tokyo Olympics	7,023	93% in July 2025	Opening ceremony 23 July 2021
Covid vaccination	9,310	94% in July 2025	Rollout mid-2021

Probe	Tweets	Concentration	Real-world date
Bitcoin Pizza Day	1,273	91% in May 2025	22 May, annually
El Salvador	16,448	Peak June 2025	Bitcoin Law announced June 2021
Beijing	326	Cluster February 2026	Winter Olympics February 2022

Everyone is out by exactly four years, with the month and day intact. The halving probe is the one that does not concentrate, sitting between 150 and 611 tweets in every month, which is expected rather than contradictory since the halving is discussed continuously rather than on a date. The tweets confirm it directly: one of the manually labelled validation tweets quotes Wozniak calling Bitcoin “pure gold now in 2022”, and another refers to a fall from $44,000, which happened in January 2022.

The dataset is therefore a re-dated copy of a 2021 to 2022 corpus, and the true coverage is 6 March 2021 to 2 March 2022. The timestamps are internally consistent, which is the most recoverable of the possible outcomes: the correction is to subtract four years from the daily index before the price merge.

Question 1 is unaffected because topic models depend on the ordering of the calendar rather than its absolute values, and the ordering is intact. Question 2 is affected, and this run does not correct it. The price series was downloaded for 2025 to 2026 while the tweets are really from 2021 to 2022, so the merge pairs sentiment with prices four years away from it. Every result in section 5 should therefore be read as a demonstration that the pipeline runs correctly rather than as a finding about Bitcoin. This is stated here rather than buried, because a null result on misaligned data is not evidence that sentiment is uninformative.

7. Conclusion and limitations

Topic modelling recovered ten themes from 199,992 tweets, with the number selected from a coherence curve that peaks cleanly at ten. The most useful result is not the themes themselves, but what they revealed about the corpus: five of the ten are promotional templates rather than conversation, accounting for 18.1% of the sample, and one further topic absorbs 49.4% as a residual. Bitcoin Twitter over this period is substantially automated and promotional, and a topic model makes that visible in a way a volume count does not.

Sentiment does not improve volatility forecasting in this run. Three pieces of evidence agree: the sentiment block fails its joint F-test at $p = 0.373$ while worsening both information criteria, no individual sentiment coefficient is significant, and in the backtest sentiment makes OLS worse and random forest unchanged, with only XGBoost improving toward rather than past the baseline. Directional accuracy sits at or below 50% for all six models. What predicts next-day volatility is market structure, specifically yesterday's return and yesterday's trading range.

There are good reasons to expect that null even on correctly aligned data. The sentiment series barely moves, varying by under 12% of its own level across the period. CryptoBERT

labels only 6.0% of tweets bearish, so there is little negative signal to work with. And the manual validation put CryptoBERT at 48.3% against human labels, so the input series is itself about half wrong, which caps how much it could ever explain.

Four limitations qualify everything above.

First, the dataset is misdated by four years. Question 2 pairs 2021 to 2022 tweets with 2025 to 2026 prices. The volatility results demonstrate that the pipeline works, rather than establishing anything about Bitcoin. Correcting it is a single line, and it is the first thing to do if this is carried further.

Second, only 201 of roughly 360 calendar days carry tweets, and daily volume ranges from 296 to 49,036, so daily sentiment is estimated from very different sample sizes without being weighted for that.

Third, the promotional templates that occupy half the Question 1 topics remain in the sentiment corpus, and both methods score them as bullish enthusiasm, biasing the series upward for reasons unrelated to market sentiment.

Fourth, the topic sample of 199,992 is a compute constraint. It recovers dominant themes precisely but loses the tail, so rare topics may be absent rather than non-existent.

On causality, none of this would support a causal claim even with correct dates. Sentiment and volatility both respond to news, so bearish tweets followed by a volatile day are at least as consistent with both reacting to the same event as with tweets moving the market. The lag structure rules out contemporaneous leakage but not confounding. The evidence here supports association at most, and in this run it does not support even that.

Three changes would make the biggest difference to a second pass. Correct the four-year offset, which costs one line. Add a template filter keyed on the first eight tokens rather than the whole string, which would clear the promotional topics and probably split the residual topic into something interpretable. And weight daily sentiment by the number of tweets behind it, so thin days stop contributing the same noise as well-covered ones.

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